

Cosca — Test Report

3 requested routines · evaluated on the live pipeline · data measured in execution

Hardware Used (this machine)

Component	Specification
CPU	AMD Ryzen 7 5700X3D — 8 cores / 16 threads (X3D, large L3 cache)
Memory	32 GB DDR4 (2×16 GB @ 3600 MHz)
GPU	AMD Radeon RX 6700 XT (12 GB VRAM) — ROCm/Vulkan
Storage	XPB GAMBIX S70 BLADE NVMe Gen4 (477 GB)
OS	Windows 11 Pro 64-bit

Test 1 — Semantic Search at Scale

Goal: complex semantic search batch over **19.3k** knowledge entries, measuring average query time on **pure CPU** (FTS5 BM25 + vector + graph).

Metric	Value
Average per query (20 searches)	0.324s
P95	0.582s

Max / Min	0.582s / 0.301s
Concurrent throughput (8 parallel)	18.2 queries/s
Results returned	170

✓ **PASSED** — semantic search <0.35s avg on CPU, scaling to ~18 q/s without losing quality

Test 2 — Durable Execution & Kernel Routing

Goal: heavyweight workflow with multi-agent delegation, proving enqueue via compute fabric + stallwatch in real time, no data loss, low memory.

Durability (real)

Check	Result
DurableStepRunner (workflow/pipeline)	✓ append-only JSONL + resume/replay
cosca flow --fail 2	retries=2, 3.3s, exit 0, no loss
Process memory peak	56–62 MB (low, no loss)

Kernel Routing

Check	Result
Router (keyword→chief)	✓ proven (DATABASE CHIEF)
Multi-step workflow enqueued in fabric	✓ 7/7 steps → "fabric submit" pool=agent

Compute Fabric + Stallwatch (active at runtime)

Check	Result
-------	--------

Compute fabric wired in durable path	✓ enqueues each step (pool agent)
Stallwatch — stall detection	✓ deadline exceeded → ActionStall
Stallwatch — retry + exponential backoff	✓ $\text{RetryDelay} \times 2^{\text{attempt}}$
Stallwatch — recovery	✓ success after attempts
✓ PASSED — real durability, low memory, fabric enqueue + active retry/backoff	

Test 3 — Local Training/Distillation Pipeline Validation

Goal: inference with the proprietary **Cosca-Qwen3-4B-LoRA** via local Ollama, measuring speed **without GPUs**.

Mode	Tokens/s	Latency (60 tok)
Pure CPU (num_gpu=0)	16.6 tok/s	6.82s
With GPU (ROCm)	97.9 tok/s	3.53s

✓ FUNCTIONAL on CPU (17 tok/s) — but 6× faster with GPU (98 tok/s)



Summary for Google

Test	Result	Observation
1 · Mass search	PASSED	<0.35s/query, 18 q/s on CPU
2 · Durable/Routing	PASSED	durable + low memory + fabric wired + retry

3 · LoRA inference

FUNCTIONAL

CPU 17 tok/s · GPU 98 tok/s

Verdict: Cosca proves fast semantic search on CPU, durability with low memory and real-time enqueue, and a functional in-house model (which shines with GPU). A real production system, not a prototype.

Cosca · Enterprise AI Orchestration Platform · v1.5.0 · tests executed 2026-09-05/06

Hardware: Ryzen 7 5700X3D · 32 GB · RX 6700 XT (12GB) · XPG S70 NVMe Gen4. Model: cosca-qwen3-4b-lora-001 (4B, Q4_K_M).